Definition 1. An explanation of the meaning of a mathematical word or phrase is called a **definition**.

A statement believed to be true and has yet to be proved is called **conjecture**.

A statement that is true and has been proved to be true is named

- **Theorem:** if it is the major result or has relevant importance in the topic,
- **Proposition:** if it has less relevance,
- **Lemma:** if it is a technical statement,
- **Corollary:** if it is a consequence of a previously stated theorem.

Given a statement P, to prove P is to show beyond a doubt that P is true. To disprove P, is to show that $\sim P$ is true.

Our goal this week is to either prove or disprove conditional statements, i.e., we will prove that one, and only one, of the following statements are true:

$$P \implies Q, \text{ or } \sim (P \implies Q).$$

Example 1. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Solution. Suppose that $A \subseteq B$ and $B \subseteq C$. Then every element a of A is an element of B, and every element b of B is an element of C. Thus every element of a of A is an element of C. Therefore $A \subseteq C$.

Example 2. Let A and B be subsets of a set U. If $A \subset \overline{B}$, then $A \cap B = \emptyset$.

Solution. Suppose that $A \subset \overline{B}$. Then every element a of A belongs to the complement of B. Thus, a does not belong to B, and in particular, does not belong to $A \cap B$. Thus, $A \cap B$ must be empty.

Definition 2. An integer number n is **even** if $n = 2a$ for some integer $a \in \mathbb{Z}$.

Definition 3. An integer number n is **odd** if $n = 2a + 1$ for some integer $a \in \mathbb{Z}$.

Definition 4. Two integers x, y have the same parity if they are both odd or both even. Otherwise, we say they have opposite parity.

Example 3. If x and y are odd integers, then xy is odd.

Solution. $xy = (2a + 1)(2b + 1) = 4ab + 2a + 2b + 1 = 2(2ab + a + b) + 1$

Definition 5. Suppose a and b are integers. We say that a **divides** b, written $a \mid b$, if $b = ac$ for some $c \in \mathbb{Z}$. In this case, we also say that a is a **divisor** of b, and that b is a **multiple** of a.

Example 4. Let a, b and c be integers. If $a \mid b$ and $b \mid c$, then $a \mid c$.

Solution. Suppose $a \mid b$ and $b \mid c$. Then $b = an$ for some $n \in \mathbb{Z}$, and $c = bm$ for some $m \in \mathbb{Z}$. Thus, $c = bm = (an)m = a(nm)$, so $c = ax$ for the integer $x = nm$. Therefore $a \mid c$.

Example 5. Suppose a, b, c and d are integers. If $a \mid b$ and $c \mid d$, then $ac \mid bd$

Solution. Suppose $a \mid b$ and $c \mid d$. Then by definition, $b = an$ for some $n \in \mathbb{Z}$. Likewise, $d = cm$ for some $m \in \mathbb{Z}$. Thus, $bd = (an)(cm) = (ac)(nm) = (ac)x$ for the integer $x = nm$. Therefore, $ac \mid bd$.

Definition 6. A number $p \in \mathbb{N}$ is **prime** if it has exactly two positive divisors, 1 and p. If it has more than two positive divisors, it is called **composite**.

Example 6. For any $n \in \mathbb{N}$, if p is prime, then the divisors of p^n are $1, p, p^2, \ldots, p^{n-1}$. (Just state for $n = 2$)

Example 7. Suppose that p is prime. If p^2 is even, then $p = 2$.

Solution. Suppose p^2 is even. Then $p^2 = 2a$ for some $a \in \mathbb{Z}$. Then $2 \in \{1, p, p^2\}$. Since 2 can't be 1 nor p^2, it follows that $p = 2$.

Example 8. Suppose that p is a positive integer greater or equal to two. If $p^2 + 1$ is prime, then $p = 2$ or p is not prime.

Solution. Suppose that $p^2 + 1$ is prime. Since $p \geq 2$, it follows that $p^2 + 1 \geq 5$. Thus, $p^2 + 1$ must be an odd prime, that is, $p^2 + 1 = 2a + 1$ for some $a \in \mathbb{Z}$, so $p^2 = 2a$, and thus p is even. So $p = 2$ or p is not prime.

Example 9. If $n \in \mathbb{Z}$, then $5n^2 + 3n + 7$ is odd.

Solution. Split into different cases.

Example 10. If $n \in \mathbb{Z}$, then $n^2 + 3n + 4$ is even.

Solution. Split into different cases.

Example 11. Let x and y be positive numbers. If $x \leq y$, then $\sqrt{x} \leq \sqrt{y}$.

Solution. Suppose $x \leq y$, so $x - y \leq 0$. Note that

$$0 \geq x - y = (\sqrt{x})^2 - (\sqrt{y})^2) = (\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y})$$

so dividing by $(\sqrt{x} + \sqrt{y})$ on both sides yields $0 \geq \sqrt{x} - \sqrt{y}$, which is $\sqrt{y} \geq \sqrt{x}$.

Example 12. If x and y are positive real numbers, then $2\sqrt{xy} \leq x + y$.

Solution. Suppose x, y are positive real numbers. Note that $0 \leq (x-y)^2 = x^2 - 2xy + y^2$, adding $4xy$ on each side yields
$$4xy \leq x^2 + 2xy + y^2 = (x+y)^2,$$
so taking the square root on both sides proves the result.

Example 13. Prove that $f(x) = x$ is continuous at $a = 0$, i.e., prove that $\forall \varepsilon > 0, \exists \delta > 0, |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon$.

Solution. $\delta = \varepsilon$

Example 14. Prove that $f(x) = x^2$ is continuous at $a = 0$, i.e., prove that $\forall \varepsilon > 0, \exists \delta > 0, |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon$.

Solution. $\delta = \sqrt{\varepsilon}$

Definition 7. The **least common multiple** of non-zero integers a and b, denoted $\operatorname{lcm}(a, b)$, is the smallest integer in $\mathbb{N}$ that is multiple of both a and b.

Definition 8. The **greatest common divisor** of integers a and b, denoted $\gcd(a, b)$, is the largest integer that divides both a and b.

Example 15. Let p, q be prime numbers. If $\gcd(p, q) > 1$, then $p = q$.

Solution. Suppose $n = \gcd(p, q) > 1$. Note that $n \mid p$, so $n = 1$ or $n = p$. Likewise, $n = 1$ or $n = q$. Since $n > 1$, it follows that $n = p$ and $n = q$. Thus, $p = q$.

Definition 9. Two integers a and b are said to be relatively prime if $\gcd(a, b) = 1$. (or coprimes)

Example 16. If a, b, c are positive integers, then $lcm(ac, bc) = c \cdot lcm(a, b)$.

Solution. Let a, b, c be positive integers, and let $m = lcm(ac, bc)$ and $n = c \cdot lcm(a, b)$.

On the one hand, $lcm(a, b) = ax = by$ for some $x, y \in \mathbb{Z}$. So $n = acx = bcy$ is a multiple of both ac and bc. Since m is the smallest such number, $m \leq n$.

On the other hand, $m = acx = bcy$ for some $x, y \in \mathbb{Z}$. So $\frac{m}{c} = ax = bc$ is a multiple of both a, b. Since $\frac{n}{c}$ is the smallest such number, $\frac{n}{c} \leq \frac{m}{c}$, i.e., $n \leq m$.

Therefore, we've shown that $n \leq m$ and $m \leq n$, hence $n = m$.

Counterexamples

The idea is to prove that the negation of "If P, then Q" is true, i.e., "P and not Q". Analogously, "for all x such that P, then Q" becomes "there exists x such that P and not Q".

Example 17. For every $n \in \mathbb{N}$, the integer $2^n - 1$ is prime. $n = 11 \implies 2^n - 1 = 2047 = 23 \cdot 89$

Example 18. If A, B and C are sets, then $A - (B \cap C) = (A - B) \cap (A - C)$. $A = \{1, 2, 3\}$, $B = \{1, 2\}$, $C = \{2, 3\}$.

Example 19. If A, B and C are sets, then $A - (B \cup C) = (A - B) \cup (A - C)$. $A = \{1, 2, 3\}$, $B = \{1, 2\}$, $C = \{2, 3\}$.

Example 20. If $x, y \in \mathbb{R}$, then $|x + y| = |x| + |y|$. $x = 1$, $y = -1$

Example 21. If $n \in \mathbb{Z}$ and $n^5 - n$ is even, then n is even. $n = 1$

Example 22. If $a, b \in \mathbb{Z}$, $a \mid b$ and $b \mid a$, then $a = b$. $a = 1$, $b = -1$

Example 23. For any irrational numbers a and b, the number a^b is irrational. $a = \sqrt{2}^{\sqrt{2}}$, $b = \sqrt{2}$

Example 24. All functions $f : \mathbb{R} \to \mathbb{R}$ are continuous.